

Cloud Network Design & Migration: Securing Crestline Bank

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Target Goal: High-Availability Financial Network Modernization

The Mission: Dragging Core Banking out of the Danger Zone

Before this modernization initiative, Crestline Bank's core banking applications lived on a single, vulnerable on-premises server. This legacy setup suffered from zero structural redundancy, no internal network isolation, and high-risk, direct exposure to the public internet. If that single hardware node failed or was compromised, banking operations would instantly grind to a halt.

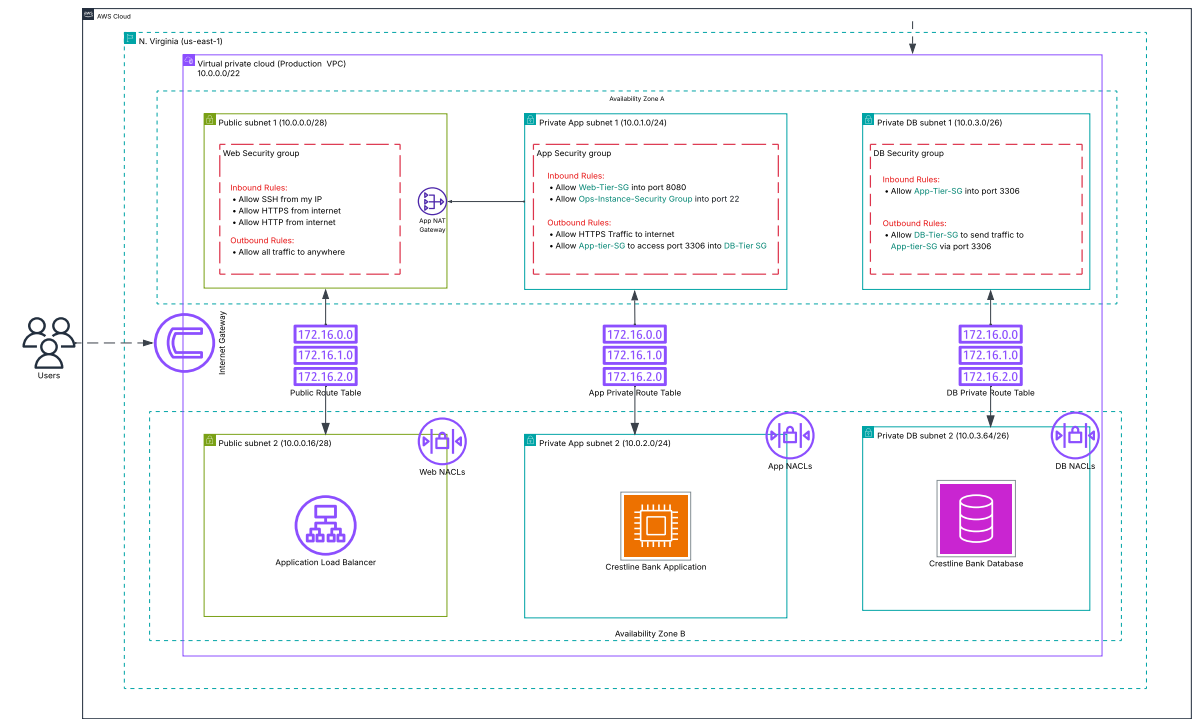
The Ironclad Core Directive:

"No internal backend system, application logic tier, or ledger database should ever be directly reachable from, or exposed to, the public internet."

The ultimate goal of this deployment was to completely re-engineer the bank's digital perimeter from scratch, ensuring that customer assets and financial transaction streams are permanently locked behind an elite, multi-layered cloud security fortress.

The Architecture Blueprint: Strategic Network Zoning

Instead of throwing every server into one open network room, we split the architecture into separate, specialized security zones using an advanced AWS Virtual Private Cloud (VPC) network footprint spanning a massive pool of **1,024 highly secure IP channels**. To guarantee that a physical power outage or localized data centre disaster never takes down the bank, the network is mirrored evenly across two distinct geographic datacentres (**Availability Zones 1 & 2**).

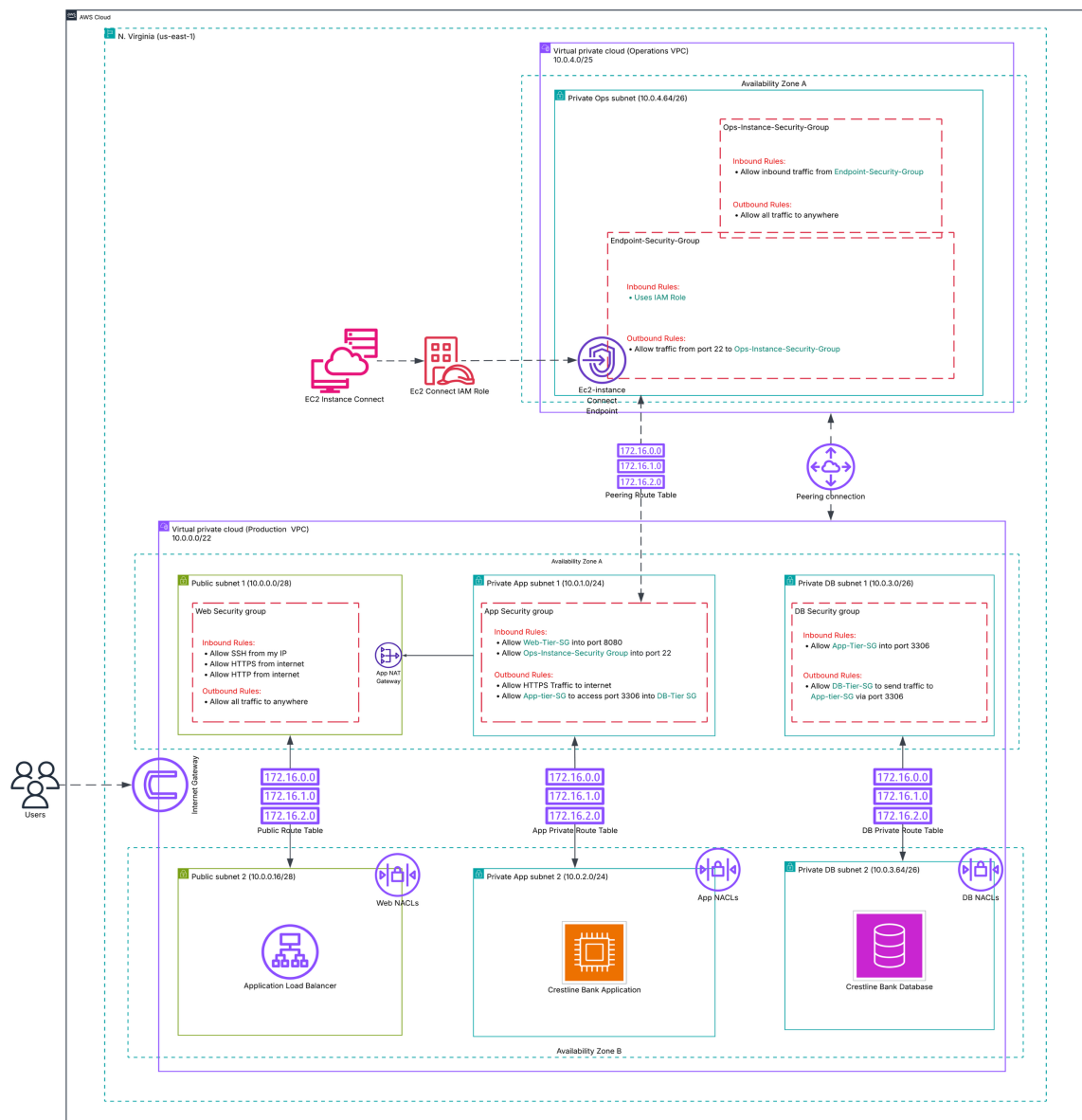


The Production Isolation Tiers

- **The Public Entry Gateways (AZ1/AZ2 Public Subnets):** Tiny, highly restricted entry points that host *only* the internet-facing Application Load Balancers (ALBs) and NAT Gateways. Absolutely no application code or data resides here.
- **The Private Application Engine (AZ1/AZ2 Private App Subnets):** The central computational engine room housing our core banking software and containerized microservices clusters. It is completely invisible to the public eye.
- **The Isolated Database Vault (AZ1/AZ2 Private DB Subnets):** The deep interior tier containing the bank's primary transactional ledgers and relational data stores. This tier maintains **absolute internet isolation**—it holds zero network pathways to or from the outside world.

Layered Defence-in-Depth: Security Groups & Perimeter Chaining

We implemented a strict "**Defence-in-Depth**" security model. This approach means an intruder cannot breach the system by breaking a single lock; instead, they face multiple, independent, automated security barriers at every layer of traversal.



Our security boundaries utilize an advanced mechanism called **Security Group Chaining**, which validates traffic dynamically based on identity rather than arbitrary IP lists:

1. **The Web Layer (SG-Web-Tier)** accepts public web traffic on ports 80 and 443. However, legacy administrative access (SSH Port 22) is completely shut down from the public and locked exclusively to the manager's authorized IP address.
2. **The App Layer (SG-App-Tier)** completely ignores the public internet. It will only open its gates if an incoming request originates directly from the trusted SG-Web-Tier on port 8080.
3. **The Database Layer (SG-DB-Tier)** is completely deaf to the outside world. It exclusively processes SQL queries arriving directly from the validated SG-App-Tier on port 3306. If any other node attempts an access request, the connection instantly times out.

Seamless Operations & Private Cross-VPC Peering

To maintain a banking network that has no direct access to the internet, we built an independent, isolated **Operations VPC network** (10.0.4.0/25) to run out-of-band management tools.

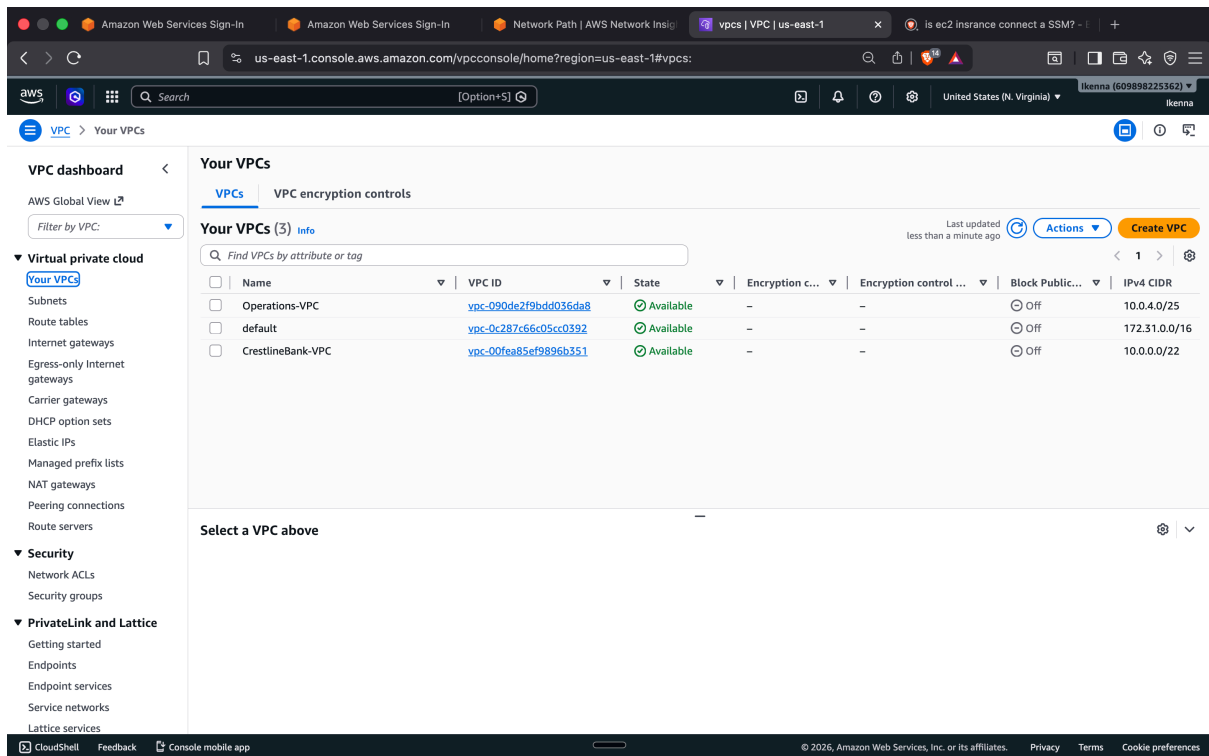
The screenshot displays two AWS Management Console pages. The top page shows the EC2 Instances list, and the bottom page shows the VPC Security Groups list.

EC2 Instances (4)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
Web-Tier-Cres...	i-06702077e2676499	Running	t3.micro	3/3 checks passed	us-east-1a	-	13.220.183.185	-	-
DB-Tier-Crestl...	i-0e5151c8155c3952b	Running	t3.micro	3/3 checks passed	us-east-1a	-	-	-	-
App-Tier-Crest...	i-088c753926f0bb728	Running	t3.micro	3/3 checks passed	us-east-1a	-	-	-	-
OpsInstance	i-0744bd06efdea5bbe	Running	t3.micro	3/3 checks passed	us-east-1a	-	-	-	-

Security Groups (8)

Name	Security group ID	Security group name	VPC ID	Description
-	sg-092d9601f08869195	default	vpc-0c287c66c05cc0392	default VPC security group
-	sg-0aad573aad2f5bd7	default	vpc-090de2f9bdd036da8	default VPC security group
App-SG-Crestline-Bank	sg-0960fe3616d48d622	App-SG-Crestline-Bank	vpc-00fea85ef9896b351	Port 8080 from SG-Public-Tier
Operations-SG-Crest...	sg-0a7d0c78befb9cba9	Operations-SG-Crestline-Bank	vpc-090de2f9bdd036da8	Operations Security Group for
DB-SG-Crestline-Bank	sg-0bbfefeefed8e53f6	DB-SG-Crestline-Bank	vpc-00fea85ef9896b351	Database Security Group
-	sg-082d1c802a4e9a1ce	ec2Instance-connect-endpoint-sg	vpc-090de2f9bdd036da8	EIC Endpoint Security Group
Web-SG-Crestline-Ba...	sg-04a25e549dd6d3e2c	Crestline-Bank-Public-SG	vpc-00fea85ef9896b351	Public Instance Security Group
-	sg-005879f7f96467030	default	vpc-00fea85ef9896b351	default VPC security group



- **Zero Public Entry points (SSM Session Manager):** Engineers log into the management cluster using a secure session token system that routes through private endpoints. No public-facing passwords or open internet entry points are required.
- **The Invisible Highway (VPC Peering):** Management traffic crosses between the Operations VPC and the Production VPC over an encrypted **VPC Peering Connection**. This architecture routes packets directly across AWS's internal private fibre backbone, completely bypassing the hazards of the public internet.

Automated Validation: The Proof of Isolation Matrix

Before deploying a single production financial service, we subjected this network layout to aggressive, real-world penetration and isolation simulation testing.

The system achieved a flawless validation run:

Sequence	Real-World Simulation Challenge	Target Tier	Expected & Actual Result	Security Capability Proven
1	External Ingress Validation	Web Subnet	SUCCESS	Public consumers can reach the load balancers smoothly.

2	Internal Multi-Tier Jump	App Subnet	SUCCESS	The web tier safely chains and offloads requests to the application core.
3	Secure App Outbound Patching	App Subnet	SUCCESS	App servers can pull vital security patches via a protected outward gate.
4	Direct Public External Cyber Attack	App Subnet	FAIL / TIMEOUT	True private shielding; external threats cannot see or route to internal systems.
5	Database Internet Leak Test	DB Subnet	FAIL / TIMEOUT	Database is 100% blind to the internet, meeting strict regulatory data audits.
6	Secure Cross-Network Administration	App Subnet	SUCCESS	Authorized engineers can smoothly navigate across our private peering highway.
7	Internal Database Lateral Attack	DB Subnet	FAIL / TIMEOUT	Even if the management tier is compromised, the database drops non-app queries.

Executive Compliance & Audit Sign-off

- **Zero-Leak Database Zone:** Confirmed that core ledger components maintain absolute isolation from the outside world.
- **No Single Point of Failure (SPOF):** Multi-AZ deployment ensures that if an entire datacentre suffers a physical failure, the companion zone automatically handles all traffic.
- **Regulatory Ready:** Fully compliant with cloud design benchmarks required for modern financial tech applications.